



INTERNET OF THINGS

Z-Wave or BACnet can connect to the Alljoyn network easily.

Thread. Thread protocol is the brainchild of Thread Group, whose founding members are Yale Security, Silicon Labs, Samsung Electronics, Nest Labs, Freescale Semiconductor, Big Ass Fans and ARM. It is primarily a creation of Google NEST group and is an Internet Protocol (IP) based solution built on open standards and designed for low-power mesh networks.

The protocol for Thread is framed with a low power angle in mind and is built on well-proven

industrial approaches to standardise IoT communications and interoperability between connected devices, with its diamond members being Cisco, GE Software, Intel, Mediatek and Samsung.

IoTivity framework application program interfaces (APIs) work on a resource based RESTful architecture model operating as middleware for connected systems, while data transmission is based on a messaging and streaming model.

The compatibility angle

All three standards aim at working

home automation. Along with this, Alljoyn supports developing devices in the fields of smart television, smartaudio, broadband gateways and automotive, while smartcity and wearables are the focus areas for IoTivity. Thread adds focus on environmental IoT devices, too. For all devices to connect to these networks and be used, these need to undergo a stringent standard compliance and certification by their respective organisations.

One main concern with the IoT these days is the security aspect. Keeping your personal data from prying eyes involves multiple layers of encryption and strict authentication. All three standards follow this to the T. The latest version of Alljoyn even includes additional features like elliptical curve cryptography and access-control lists to track device nodes.

On being asked about standards, in an interview with EFY, Siddharth Unny, director, TL Micro Automation, had very interesting views. With Europe being the leader in standardisation today, it is terribly costly for Indian products to be certified. “Why not have an exclusive standard for India, one that suits our requirements?” he asked.

Having a standard that is not as stringent as European allows Indian players to provide low-cost products and get these to market, quicker. This helps them compete with big players who price their product through economies of scale.

He also went on to add that one of the best ways to keep data secure would be to use your own custom protocols. This could be just for your dedicated network, that is, till the gateway. Apart from keeping your data secure, this provides the flexibility to design what you want and how, within your constraints.

What do we have apart from these three

No discussion on the IoT is com-

“The IoT is all about heterogeneous and aware devices interacting to simplify people’s lives in one way or the other. The new paradigm of the IoT has given rise to a new way of communication between humans, machines and things. Bringing the IoT to life requires a comprehensive systems approach, inclusive of intelligent processing and sensing technology, connectivity, software and services, along with a leading ecosystem of partners.

So what can a developer do? Support all standards? Too expensive. Wait for one standard to replace all others? May not happen. Implement a software abstraction layer that permits certain interfaces/standards to be replaced? Good, if possible, but not always.

Choose one standard and accept incompatibility with all others?

Bad, but sometimes the only choice. And so on.”

—Narang N. Kishor, mentor and principal design architect, Narnix Technolabs Pvt Ltd, and convenor of panel on smart infrastructure at Bureau of Indian Standards (BIS)

standards like Institute of Electrical and Electronics Engineers (IEEE) 802.15.4, Internet Engineering Task Force (IETF) Internet Protocol version 6 (IPv6) and IPv6 over Low-power Wireless Personal Area Networks (6LoWPAN). This is an IPv6 networking protocol that operates in the 2.5GHz band at 250kbps. Thread devices create a mesh network of routers that forward messages from one node to the next.

IoTivity. This framework is hosted by Linux Foundation, sponsored by Open Interconnect Consortium (OIC) and managed by IoTivity Steering Group. OIC believes in combining the open source and the

across all platforms and operating systems (OSes), ranging from Android to iOS, Linux and Windows. The idea is to make these agnostic to all communication layers, from Wi-Fi to the Ethernet. While Thread allows about 250 devices to connect to its network at a time, Alljoyn network decides the limit by the range of the technology, available bandwidth and hardware restrictions.

One area where Thread wireless protocol differs from the other two is in leaving the application layer to be chosen by the user via User Datagram Protocol (UDP).

Keeping with the trend, all these standards seem to primarily target